

In this project, I have created a Windows virtual machine in Azure to host a new website for Sam's Scoops. I will deploy the virtual machine for the Sam's Scoops web server using the virtual network Web_Server on a 172.16.1.0 subnet. This virtual machine was named "SamScoopsWeb".

This SOP will guide a through steps as future reference for setup with screenshots that demonstrate the correct actions

Create the network and resource group

Step 1

Sign into the [Azure portal](#) with your credentials.

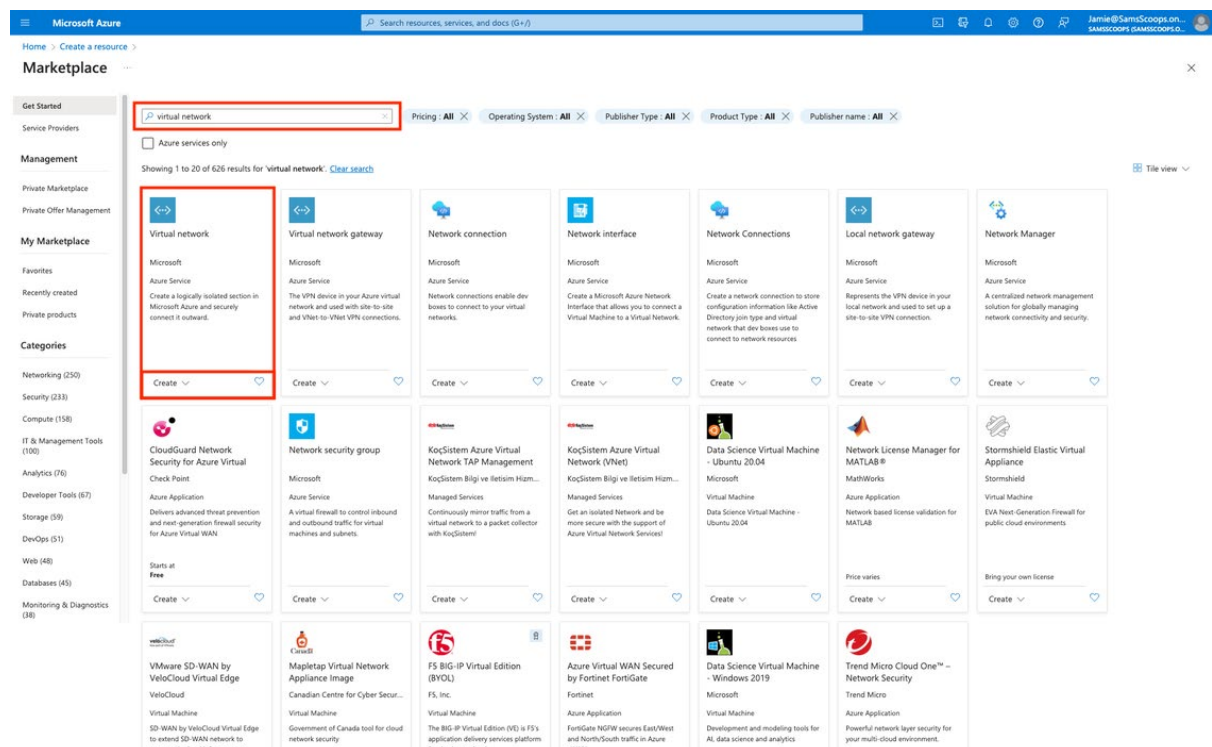
Step 2

In the Azure portal menu, select the **Create a resource** button located on the left-hand side of the screen.



Step 3

Next, type "Virtual network" in the search bar, select **Virtual network** from the results and select **Create**.



Step 4

In the **Basics** tab of the Create a virtual network wizard, fill out the following information:

1. Subscription: Choose the subscription that you want to use.
2. Resource group: **Create new** and enter "RG_Web_Server" as the name of the new resource group.
3. Name: Enter "Web_Server" as the name of the virtual network.
4. Region: Choose the region that is closest to you.

The screenshot shows the 'Create virtual network' wizard in the Microsoft Azure portal. The 'Basics' tab is selected. Under 'Project details', the 'Subscription' is set to 'Azure subscription 1' and the 'Resource group' is set to 'NetworkWatcherRG' with a 'Create new' link below it. Under 'Instance details', the 'Virtual network name' is empty and the 'Region' is set to '(US) East US'. At the bottom, there are 'Previous', 'Next', and 'Review + create' buttons.

Basics Security IP addresses Tags Review + create

Azure Virtual Network (VNet) is the fundamental building block for your private network in Azure. VNet enables many types of Azure resources, such as Azure Virtual Machines (VM), to securely communicate with each other, the internet, and on-premises networks. VNet is similar to a traditional network that you'd operate in your own data center, but brings with it additional benefits of Azure's infrastructure such as scale, availability, and isolation. [Learn more.](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

This screenshot shows a 'Create new' resource group dialog box. It contains a text input field for the 'Name' with the value 'RG_Web_Server'. Below the input field are 'OK' and 'Cancel' buttons. The dialog box is overlaid on the 'Project details' section of the 'Create virtual network' wizard, which shows the 'Subscription' as 'Azure subscription 1' and the 'Resource group' as 'NetworkWatcherRG'.

Step 5

1. Select IP addresses.

Microsoft Azure

Home > Create a resource > Marketplace >

Create virtual network

Basics Security **IP addresses** Tags Review + create

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Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Azure subscription 1

Resource group * (New) RG_Web_Server

Create new

Instance details

Virtual network name Web_Server

Region (US) East US

Deploy to an edge zone

Previous Next Review + create

2. Delete the default address space by selecting the dots (...) next to **Add a Subnet** and select **Delete Address Space**.
3. There is a warning: **You must add at least one address space to the virtual network**.
4. Select **Add an Address Space**.
5. Fill in starting address: **172.16.1.0**.
6. Fill in address space size: **/24 (256 Addresses)**.
7. Select **Add**.

Microsoft Azure

Home > Create a resource > Marketplace >

Create virtual network

Basics Security **IP addresses** Tags Review + create

Configure your virtual network address space with the IPv4 and IPv6 addresses and subnets you need. [Learn more](#)

Define the address space of your virtual network with one or more IPv4 or IPv6 address ranges. Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet. [Learn more](#)

Add an IP address space

You must add at least one address space to the virtual network.

A NAT gateway is recommended for outbound internet access from subnets. Edit the subnet to add a NAT gateway. [Learn more](#)

Add an IP address space

The address space for a virtual network has one or more non-overlapping address ranges. It is recommended to use private (RFC 1918), shared (RFC 6598), or local (RFC 4193) address ranges. [Learn more](#)

Address space type IPv4 IPv6

Starting address 172.16.1.0

Address space size /24 (256 addresses)

IP address space 172.16.1.0 - 172.16.1.255 (256 addresses)

Previous Next Review + create

Add Cancel

Step 6

Now, a Subnet has to be added.

1. Select **+ Add a subnet**.
2. Leave the details as default and select **Add**.

Microsoft Azure

Home > Create a resource > Marketplace >

Create virtual network

Basics Security IP addresses Tags Review + create

Configure your virtual network address space with the IPv4 and IPv6 addresses and subnets you need. [Learn more](#)

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[Add an IP address space](#)

Subnets	IP address range	Size	NAT gateway
	172.16.1.0 - 172.16.1.255 (256 addresses)		

[+ Add a subnet](#)

A NAT gateway is recommended for outbound internet access from subnets. Edit the subnet to add a NAT gateway. [Learn more](#)

[Previous](#) [Next](#) [Review + create](#) [Add](#) [Cancel](#)

Step 7

Select the **Review+ create** button to review the settings.

Microsoft Azure

Home > Create a resource > Marketplace >

Create virtual network

Basics Security IP addresses Tags Review + create

Configure your virtual network address space with the IPv4 and IPv6 addresses and subnets you need. [Learn more](#)

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[Add an IP address space](#)

Subnets	IP address range	Size	NAT gateway
default	172.16.1.0 - 172.16.1.255 (256 addresses)	/24 (256 addresses)	-

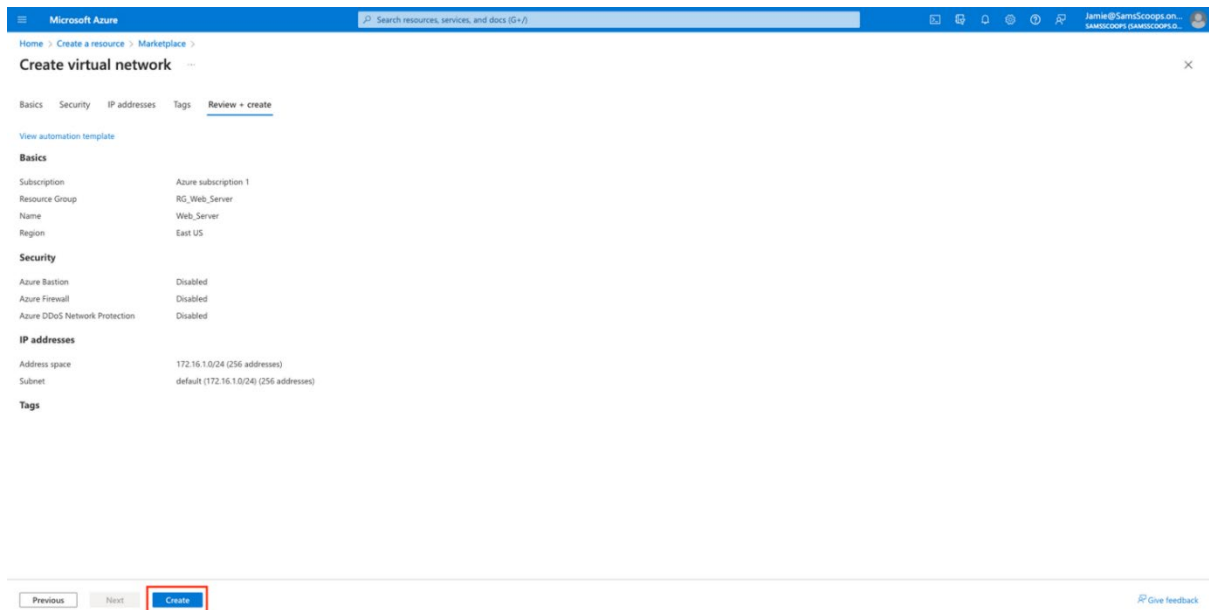
[+ Add a subnet](#)

A NAT gateway is recommended for outbound internet access from subnets. Edit the subnet to add a NAT gateway. [Learn more](#)

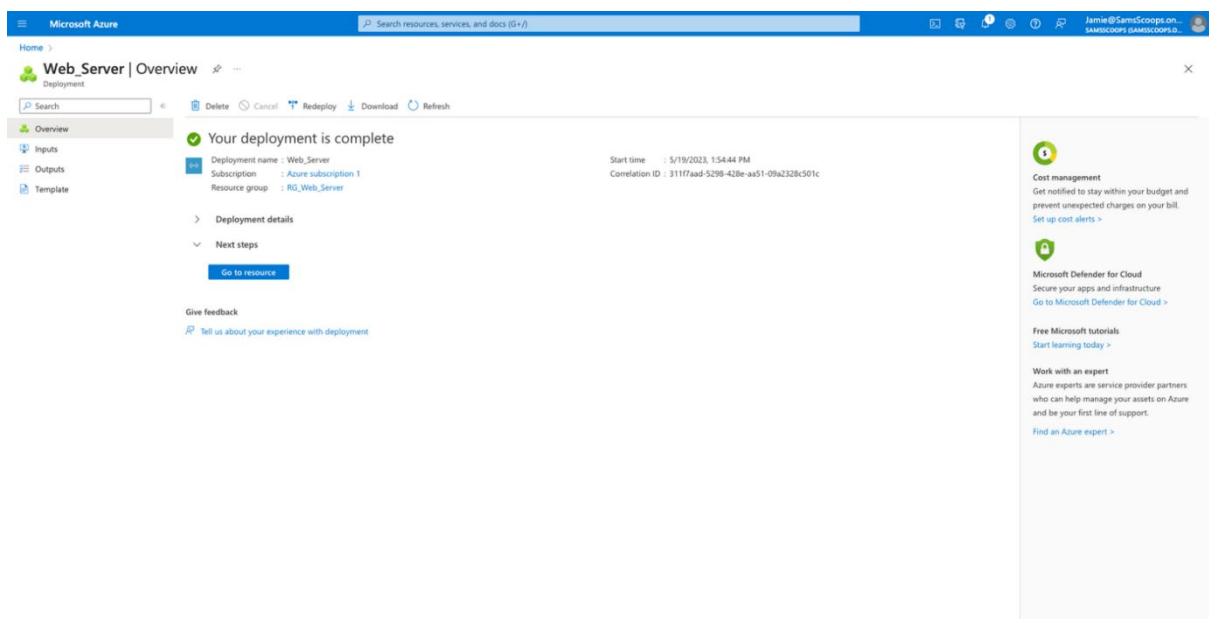
[Previous](#) [Next](#) [Review + create](#) [Give feedback](#)

Step 8

Select the **Create** button to create the virtual network.



At this stage there should be a notification that says **Deployment in progress**. When the deployment is completed, it should say **Your deployment is complete**.



Once the virtual network is created, you can proceed with creating the virtual machine for the Sam's Scoops web server.

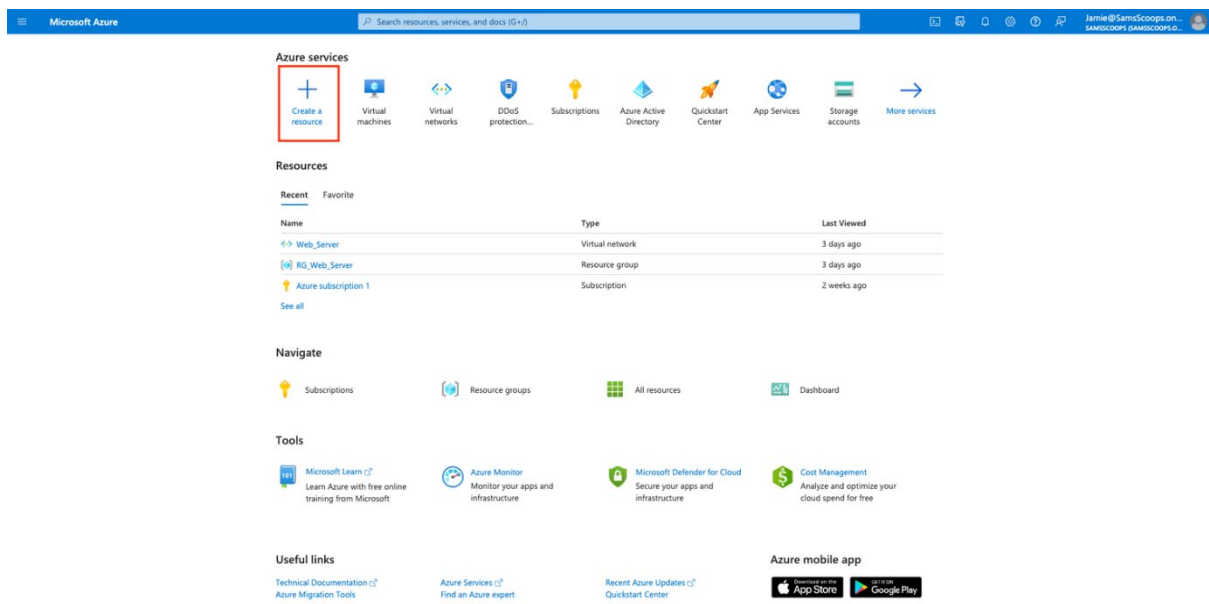
Create the virtual machine

Step 1

Sign in to the [Azure portal](#) with your credentials.

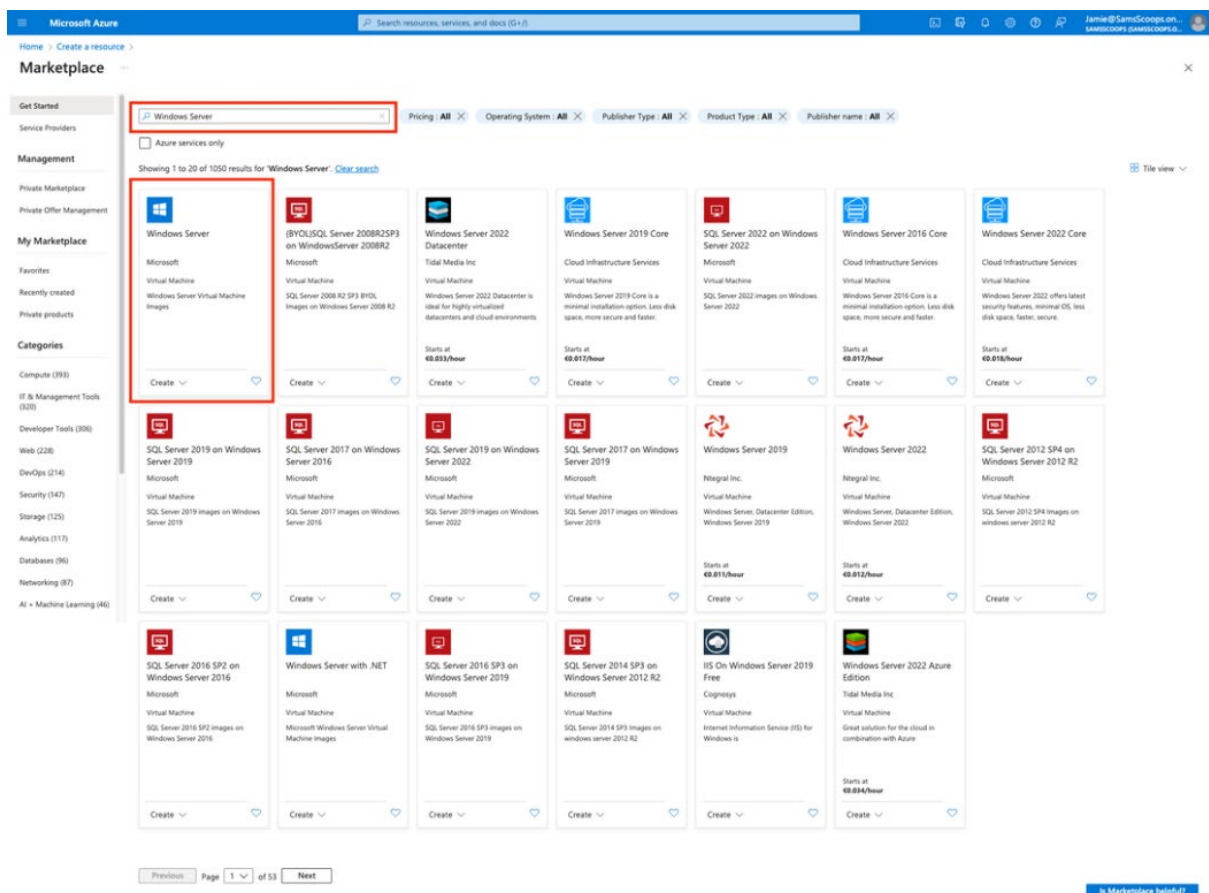
Step 2

In the Azure portal menu, select the **Create a resource** button located on the left-hand side of the screen.

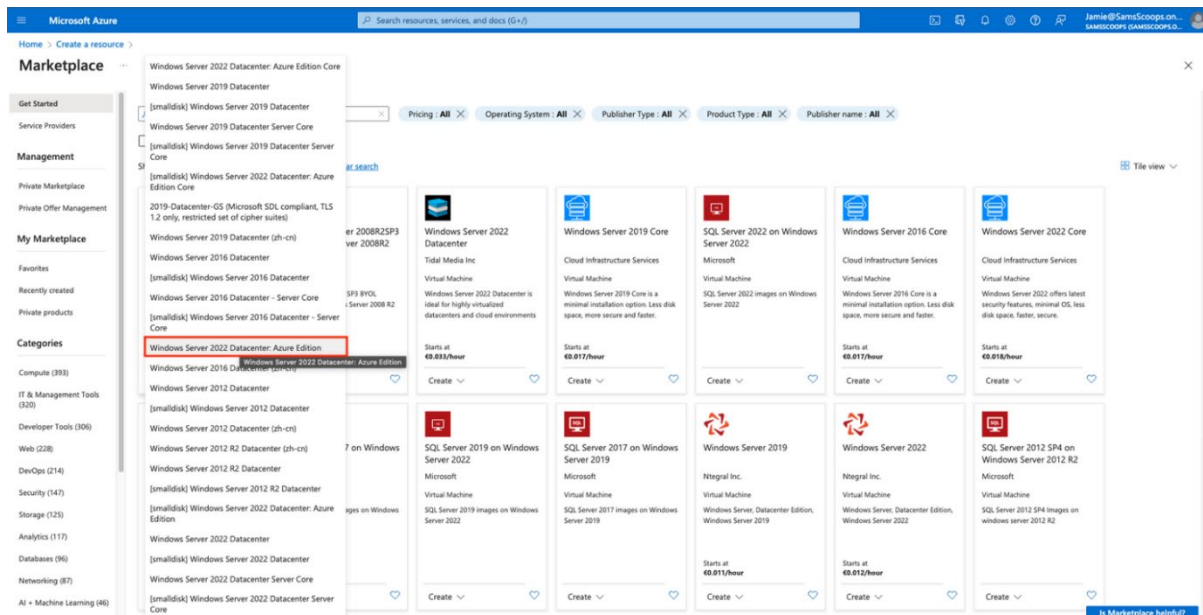


Step 3

Search for "Windows Server" in the search bar and select **Windows Server** from the results.



1. Select the **Create** and select **Windows Server 2022 Datacenter: Azure Edition**.



Step 4

In the **Basics** tab of the Create a virtual machine wizard, fill out the following information:

1. Subscription: Choose the subscription that you want to use.
2. Resource group: An existing resource group, **RG_Web_Server**.
3. Virtual machine name: Enter "SamScoopsWeb".
4. Region: Choose the region that is closest to you.
5. Image: **Windows Server 2022 Datacenter –x64 Gen2**.
6. Size: An appropriate size for your virtual machine. It will be good idea to use choose the cheapest option for this exercise.
7. Username: **AzAdmin**
8. Password: **P@\$\$@1234567**
9. Confirm Password: **P@\$\$@1234567**

Microsoft Azure

Search resources, services, and docs (5+)

Home > Create a resource > Marketplace

Create a virtual machine

Basics Disks Networking Management Monitoring Advanced Tags Review + create

Create a virtual machine that runs Linux or Windows. Select an image from Azure marketplace or use your own customized image. Complete the Basics tab then Review + create to provision a virtual machine with default parameters or review each tab for full customization. [Learn more](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription: Azure subscription 1

Resource group: RG_Web_Server

Instance details

Virtual machine name: SamScoopsWeb

Region: (US) East US

Availability options: No infrastructure redundancy required

Security type: Trusted launch virtual machines

Image: Windows Server 2022 Datacenter: Azure Edition - x64 Gen2

VM architecture: x64

Run with Azure Spot discount: ☐

Size: Standard_B1s - 1 vcpu, 1 GiB memory (\$10.22/month)

Administrator account

Username: AzAdmin

Password: [REDACTED]

Confirm password: [REDACTED]

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports: Allow selected ports

Select inbound ports: RDP (3389)

Licensing

Save up to 49% with a license you already own using Azure Hybrid Benefit. [Learn more](#)

Would you like to use an existing Windows Server license? ☐

[Review Azure hybrid benefit compliance](#)

Review + create < Previous Next > Disks

[Give feedback](#)

Step 5

Select the **Next: Disks** button to proceed to the Disks tab.

Step 6

Leave the default settings for OS disk and select the **Next: Networking** button to proceed to the **Networking** tab.

The screenshot shows the 'Create a virtual machine' wizard in the Microsoft Azure portal. The 'Disks' tab is selected, showing options for OS disk and data disks. The OS disk is set to 'Premium SSD (locally-redundant storage)'. The 'Delete with VM' checkbox is checked. The 'Key management' dropdown is set to 'Platform-managed key'. The 'Enable Ultra Disk compatibility' checkbox is unchecked. Below these options, there is a table for 'Data disks for SamScoopWeb'. The table has columns for LUN, Name, Size (GiB), Disk type, Host caching, and Delete with VM. At the bottom of the wizard, there are navigation buttons: 'Review + create', '< Previous', and 'Next: Networking >'. The 'Review + create' button is highlighted with a red box.

Step 7

In the **Networking** tab, select the following settings:

1. Virtual network: **Web_Server**.
2. Subnet: **default**.
3. Public IP: **Create new** and enter a name for the new public IP address.
4. NIC network security group: Leave the default setting.

Step 8

Select the **Next: Management** button to proceed to the Management tab.

The screenshot shows the 'Create a virtual machine' wizard in the Microsoft Azure portal, with the 'Networking' tab selected. The 'Virtual network' dropdown is set to 'Web_Server'. The 'Subnet' dropdown is set to 'default (172.16.1.0/24)'. The 'Public IP' dropdown is set to '(new) SamScoopWeb-ip'. The 'NIC network security group' dropdown is set to 'Basic'. The 'Public inbound ports' dropdown is set to 'Allow selected ports'. The 'Select inbound ports' dropdown is set to 'RDP (3389)'. Below these options, there is a warning message: 'This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.' At the bottom of the wizard, there are navigation buttons: 'Review + create', '< Previous', and 'Next: Management >'. The 'Next: Management >' button is highlighted with a red box.

Step 9

In the **Management** tab, leave the default settings and select the **Next: Monitoring** button to proceed to the **Monitoring** tab. Leave everything as default. Then select the **Next: Advanced** button to proceed to the **Advanced** tab.

The screenshot shows the 'Create a virtual machine' wizard in the Microsoft Azure portal, specifically the 'Management' tab. The 'Management' tab is highlighted in the top navigation bar. Below the navigation bar, the 'Microsoft Defender for Cloud' section is visible, indicating that the subscription is protected by the Microsoft Defender for Cloud basic plan. The 'Identity' section has a checkbox for 'Enable system assigned managed identity' which is unchecked. The 'Azure AD' section has a checkbox for 'Login with Azure AD' which is unchecked, with a note about RBAC role assignment. The 'Auto-shutdown' section has a checkbox for 'Enable auto-shutdown' which is unchecked. The 'Backup' section has a checkbox for 'Enable backup' which is unchecked. The 'Guest OS updates' section has a checkbox for 'Enable hotpatch' which is unchecked, with a note that hotpatch is not available for this image. The 'Patch orchestration options' section has a dropdown menu set to 'Automatic by OS (Windows Automatic Updates)'. At the bottom, there are three buttons: 'Review + create' (highlighted with a red box), '< Previous', and 'Next: Monitoring >' (highlighted with a red box). A 'Give feedback' link is also present in the bottom right corner.

Step 10

In the **Advanced** tab, leave the default settings and select the **Review + create** button.

Microsoft Azure

Home > Create a resource > Marketplace

Create a virtual machine

Basics Disks Networking Management Monitoring **Advanced** Tags Review + create

Add additional configuration, agents, scripts or applications via virtual machine extensions or cloud-init.

Extensions

Extensions provide post-deployment configuration and automation.

Extensions

VM applications

VM applications contain application files that are securely and reliably downloaded on your VM after deployment. In addition to the application files, an install and uninstall script are included in the application. You can easily add or remove applications on your VM after create. [Learn more](#)

[Select a VM application to install](#)

Custom data

Pass a script, configuration file, or other data into the virtual machine **while it is being provisioned**. The data will be saved on the VM in a known location. [Learn more about custom data for VMs](#)

Custom data

User data

Pass a script, configuration file, or other data that will be accessible to your applications **throughout the lifetime of the virtual machine**. Don't use user data for storing your secrets or passwords. [Learn more about user data for VMs](#)

Enable user data ☐

Performance (NVMe)

Enable capabilities to enhance the performance of your resources.

Higher remote disk storage performance with NVMe ☐

Host

Azure Dedicated Hosts allow you to provision and manage a physical server within our data centers that are dedicated to your Azure subscription. A dedicated host gives you assurance that only VMs from your subscription are on the host, flexibility to choose VMs from your subscription that will be provisioned on the host, and the control of platform maintenance at the level of the host. [Learn more](#)

Host group

Capacity reservations

Capacity reservations allow you to reserve capacity for your virtual machine needs. You get the same SLA as normal virtual machines with the security of reserving the capacity ahead of time. [Learn more](#)

Capacity reservation group

Proximity placement group

Proximity placement groups allow you to group Azure resources physically closer together in the same region. [Learn more](#)

Proximity placement group

Review + create Previous Next: Tags >

[Give feedback](#)

Step 11

Review the settings for your virtual machine and select the **Create** button to start the deployment of your virtual machine and wait for deployment.

Microsoft Azure

Search resources, services, and docs (Ctrl+K)

Home > Create a resource > Marketplace

Create a virtual machine

Validation passed

Basics Disks Networking Management Monitoring Advanced Tags **Review + create**

Cost given below is an estimate and not the final price. Please use [pricing calculator](#) for all your pricing needs.

Price

1 X Standard B1s
by Microsoft
[Terms of use](#) | [Privacy policy](#)

Subscription credits apply
0.0140 USD/hr
[Pricing for other VM sizes](#)

TERMS

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above, (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. See the [Azure Marketplace Terms](#) for additional details.

You have set RDP port(s) open to the internet. This is only recommended for testing. If you want to change this setting, go back to Basics tab.

Basics

Subscription	Azure subscription 1
Resource group	RG_Web_Server
Virtual machine name	SamScoopWeb
Region	East US
Availability options	No infrastructure redundancy required
Security type	Trusted launch virtual machines
Enable secure boot	Yes
Enable vTPM	Yes
Integrity monitoring	No
Image	Windows Server 2022 Datacenter Azure Edition - Gen2
VM architecture	x64
Size	Standard B1s (1 vcpu, 1 GB memory)
Username	JakAdmin
Public inbound ports	RDP
Already have a Windows license?	No
Azure Spot	No

Disks

OS disk type	Premium SSD LRS
Use managed disks	Yes
Delete OS disk with VM	Enabled
Ephemeral OS disk	No

Networking

Virtual network	Web_Server
Subnet	default (172.16.1.0/24)
Public IP	(new) SamScoopWeb-IP
Accelerated networking	Off
Place this virtual machine behind an existing load balancing solution?	No
Delete public IP and NIC when VM is deleted	Disabled

Management

Microsoft Defender for Cloud	Basic (free)
System assigned managed identity	Off
Login with Azure AD	Off
Auto-shutdown	Off
Backup	Disabled
Enable hotpatch	Off
Patch orchestration options	OS-orchestrated patching: patches will be installed by OS

Monitoring

Alerts	Off
Boot diagnostics	On
Enable OS guest diagnostics	Off

Advanced

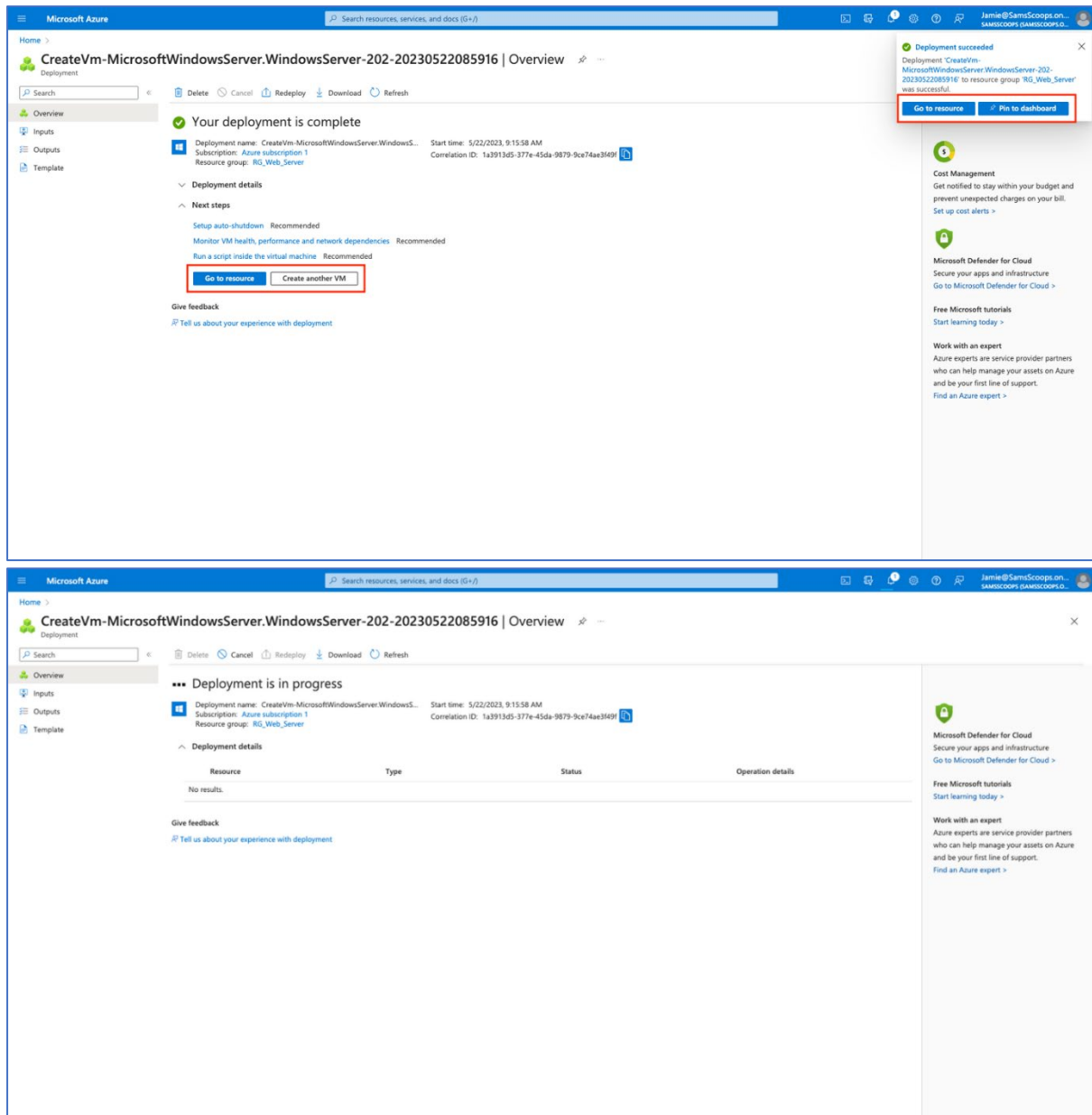
Extensions	None
VM applications	None
Cloud init	No
User data	No
Disk controller type	SCSI
Proximity placement group	None
Capacity reservation group	None

Create < Previous Next > Download a template for automation

Give feedback

Step 12

Wait for the deployment to complete.

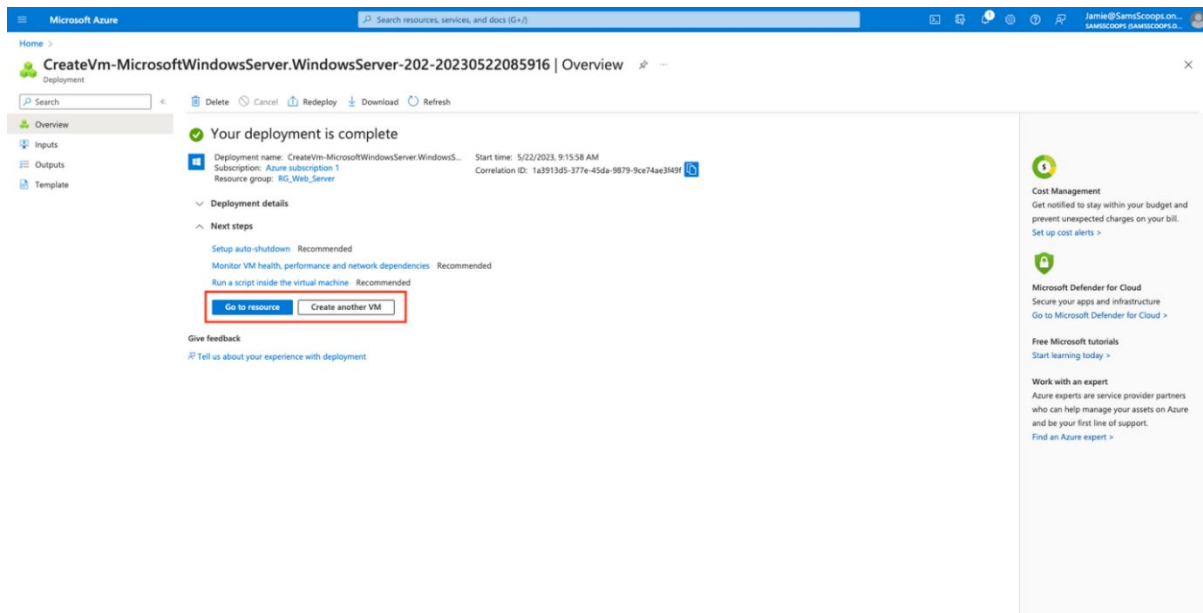


Well done. The virtual machine is now deployed. You are now ready for the next phase where you need to connect to the virtual machine.

Connect to virtual machine

Step 1

Once the deployment is complete, select the **Go to resource** button to navigate to the virtual machine page.



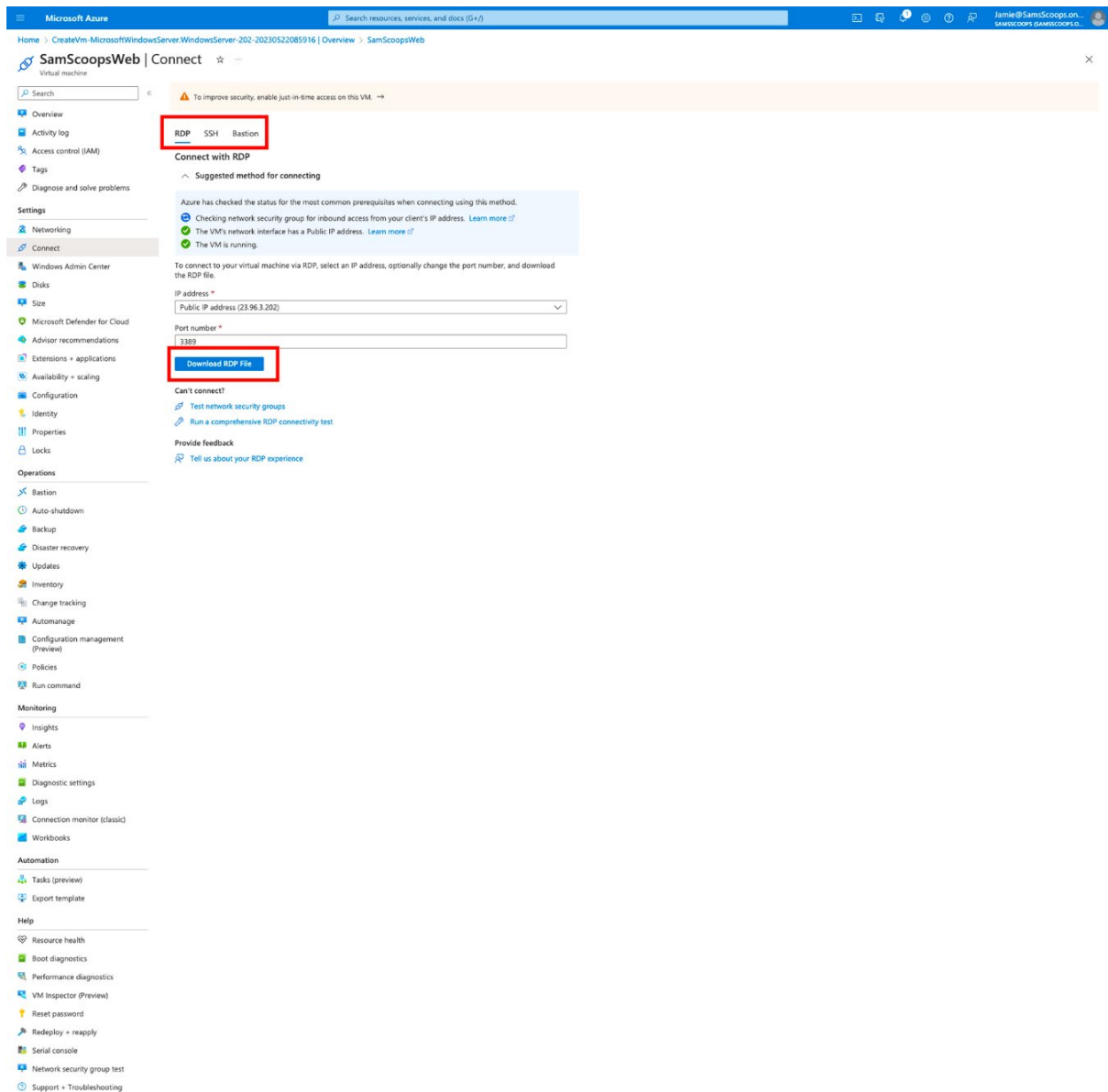
Step 2

On the virtual machine page, select the **Connect** button and **RDP** from the option menu to download the RDP file.

The screenshot shows the Microsoft Azure portal interface for a virtual machine named 'SamScoopsWeb'. The top navigation bar includes the Microsoft Azure logo, a search bar, and user information. The left-hand navigation menu lists various Azure services and tools. The main content area displays the 'Overview' tab for the 'SamScoopsWeb' VM, which is in a 'Running' state. A red box highlights the 'Connect' button in the top action bar. Below the 'Connect' button, there are several tabs: 'Essentials', 'Properties', 'Monitoring', 'Capabilities (B)', 'Recommendations', and 'Tutorials'. The 'Essentials' tab is active, showing details about the VM's resource group, status, location, subscription, and tags. The 'Properties' tab is also visible, showing details about the VM's operating system, publisher, offer, plan, VM generation, architecture, agent status, agent version, host group, host, proximity placement group, colocation status, capacity reservation group, and disk controller type. The 'Networking' tab is also visible, showing details about the VM's public IP address, private IP address, virtual network/subnet, and DNS name. The 'Size' tab is also visible, showing details about the VM's size, vCPUs, and RAM. The 'Disk' tab is also visible, showing details about the VM's OS disk, encryption at host, Azure disk encryption, and ephemeral OS disk.

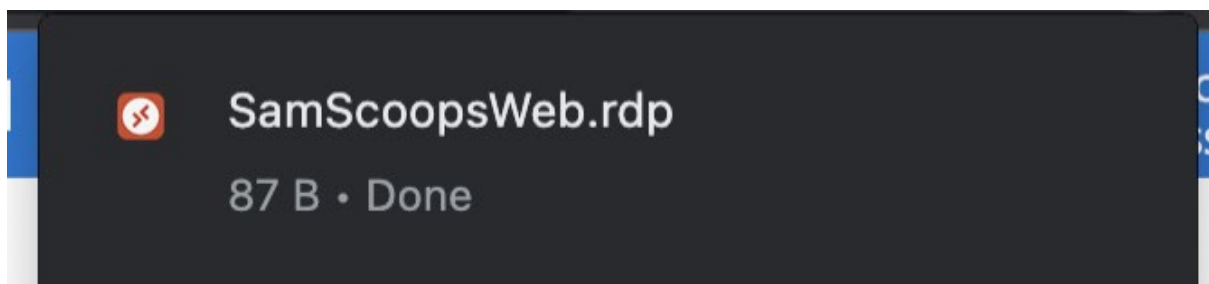
Step 3

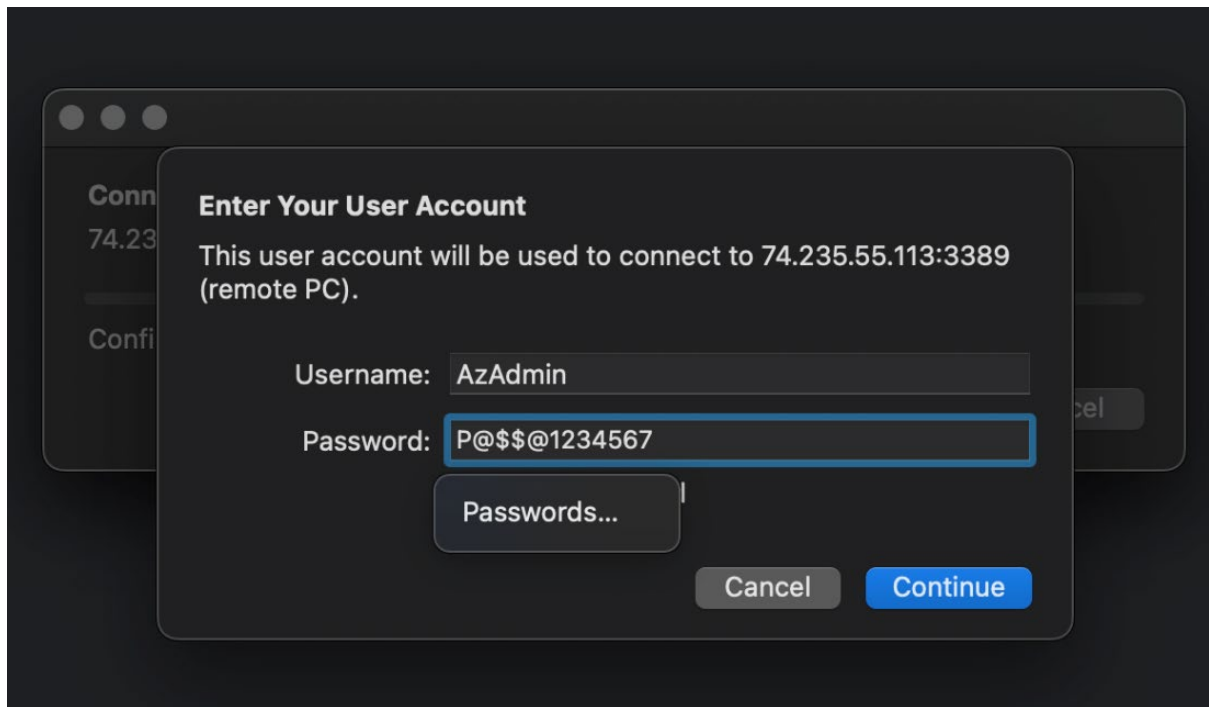
Select **RDP** from the options menu to download the RDP file. Select **Download RDP file**.



Step 4

Open the downloaded RDP file and connect by using the credentials you have set while creating the virtual machine to connect to the virtual machine.





You have now created a Windows virtual machine in the Azure portal for **SamScoopsWeb**.

